focused the sound. Unfortunately, most of the human-audible sound is a mixture of signals with varying wavelengths—between 2 cms to 17 metres (the human hearing ranges from a frequency of 20 Hz to 20,000 Hz). Hence, except for very low wavelengths, just about the entire audible spectrum tends to spread out at 360 degrees.

To create a narrow sound beam, the aperture size of the source also matters—a large loudspeaker will focus sound over a smaller area. If the source loudspeaker can be made several times bigger than the wavelength of the sound transmitted, then a finely focused beam can be created. The problem here is that this is not a very practical solution. To ensure that the shortest audible wavelengths are focused into a beam, a loudspeaker about 10 metres across is required, and to guarantee that all the audible wavelengths are focused, even bigger loudspeakers are needed.

Ultrasound to the rescue

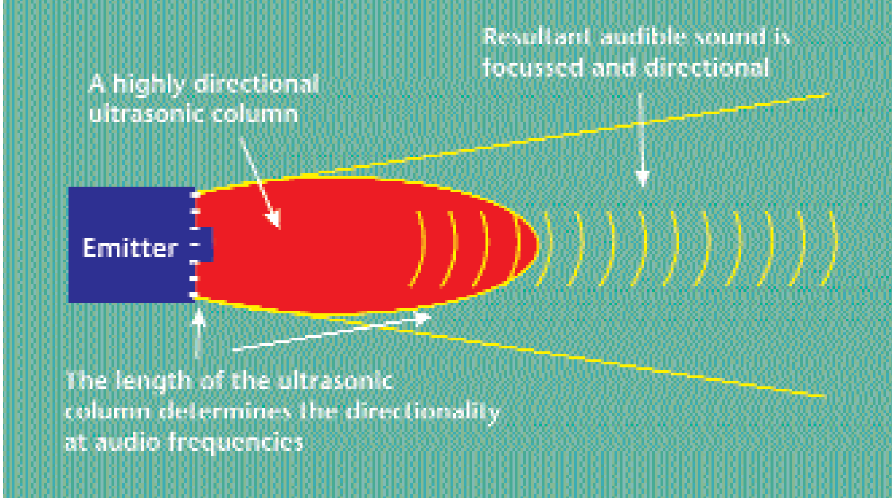
Bigger loudspeakers and source apertures seemed like an insurmountable obstacle, until scientists started work on ultrasound. Pioneering research in the usage and propagation of ultrasound can be traced back to the 1960s, when Orhan Berktay and other acousticians were studying underwater sound propagation. Researchers discovered that if short pulses of ultrasound were fired into water, the pulses were spontaneously converted into low frequency sound. Ultrasound is sound that has very small wavelength—in the millimetre range. You can't hear ultrasound since it lies beyond the threshold of human hearing. Orhan Berktay studied this and wrote his seminal paper, which established that water distorts ultrasound signals in a non-linear, but predictable, mathematical way. The researchers finally went on to use this and create directional sonar, but there was a bigger use of Berktay's equation outside of sound propagation under water. Research in the early 1970s focused on sound propagation in air. This is when researchers started firing the inaudible ultrasound pulses into the air and discovered that air spontaneously converted the inaudible ultrasound into audible sound tones, hence proving that as with water, sound propagation in air is just as non-linear, but can be calculated mathematically. Joseph Pompei's acoustic device, the Audio Spotlight works just like that—it comprises a speaker that fires ultrasound pulses, which act in a manner similar to that of a long, narrow column extending from the front of the disc. The ultrasound column acts as an airborne speaker, and as the beam moves through the air, gradual distortion takes place in a predictable way. This gives rise to audible components that can be accurately predicted and precisely controlled.

However, the problem with firing off ultrasound pulses, and having them interfere to produce audible tones is that the audible components created are nowhere similar to the complex signals in speech and music. Human speech, as well as music, contains multiple varying frequency signals, which interfere to produce sound and distortion. To generate such sound out of pure ultrasound tones is not easy. This is when teams of researchers from Ricoh and other Japanese companies got together to come up with the idea of using pure ultrasound signals as a carrier wave, and superimposing audible speech and music signals on it to create a hybrid wave. This is similar to the idea of amplitude modulation (AM), a technique used to broadcast commercial radio stations signals over a wide area. The speech and music signals are mixed with the pure ultrasound carrier wave, and the resultant hybrid wave is then broadcast. As this wave moves through the air, it creates complex distortions that give rise to two new frequency sets, one slightly higher and one slightly lower than the hybrid wave. Berktay's equation holds strong here, and these two sidebands interfere with the hybrid wave and produce two signal components, as the equation says. One is identical to the original sound wave, and the other is a badly distorted component. This is where the problem lies—the volume of the original sound wave is proportional to that of the ultrasounds, while the volume of the signal's distorted component is exponential. So, a slight increase in the volume drowns out the original sound wave as the distorted signal becomes predominant. It was at this point that all research on ultrasound as a carrier wave for an audio spotlight got bogged down in the 1980s.

Berktay's equation works

Then, Joseph Pompei realised that the focus should have been on the signal's distorted component. Now since the signal component's behavior is mathemati-

Using ultrasound to generate audible sound



Mysterious Sounds

Apart from discreet communication systems and acoustic assault rifles that the US Army is interested in, there is one really wild application—Ventriloquist systems. By using tiny sound focusing devices to beam out voices and having them scatter against rocks and natural obstacles to the path, they can give the impression of the presence of people in uninhabited places.

Tricks of this sort are known as PsyOps—a short form for Psychological Operations that are used to fight a war of wits against enemy troops.